

German embedded V2 clauses: A quasi-subordination account*

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The claim: embedded root constructions involve **extra syntactic projections** with extra **not-at-issue requirements** which is a subset of the **felicity conditions** of the corresponding speech act.
The case: German embedded V2 clauses

1 Root, embedded root, and German V2

Root clauses are different from subordinated clauses; (1) presents the classic definition. Per this definition, a clause is either root (2) or non-root (i.e., subordinated; (3)).

(1) *Root*

A root will mean either the highest S in a tree, an S immediately dominated by the highest S or the reported S in direct discourse. (Emonds 1970, 6)

- (2) a. [The earth is flat].
b. We should not travel, [because the earth is flat].
c. John says, “[the earth is flat].”
(3) John believes [that the earth is flat].

The root/subordinated distinction are sometimes formally marked. For example, English **matrix polar questions** differ from **embedded polar questions** in (i) subject-auxiliary inversion (SAI) and (ii) complementizers.¹

(4) Matrix clause: Inversion + no complementizer

(*Department chair to a new lecturer:*)

*[**Whether** you can teach Semantics next year]_↑/[**Can** you teach Semantics next year]_↑

(5) Subordinated clause: No inversion + whether

(*Secretary to the chair:*)

I know [**whether** she can teach Semantics next year]_↓/*[**can** she teach Semantics next year]_{↑/↓}

German declaratives exhibit such a root/non-root contrast in the positions of the inflected verbs. Verb-second (V2) clauses show up in matrices², whereas the verb-final word order suggests subordination.

(6) Matrix clause: V2 + no complementizer

***Dass** Peter nach Hause **geht**./ Peter **geht** nach Hause.
that Peter home towards go Peter go towards home
(Intended:) ‘Peter is going home.’

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1. Following the original notation in Dayal (2025), I use ↑ and ↓

2. I leave aside exceptions where *dass*-clauses show up in matrix contexts when certain restrictions are met. Schwabe (2006) provides a detailed distributional restrictions of such matrix *dass*-clauses, which I take to be irrelevant to the purpose of the current study.

- (7) Subordinated clause: *V-final + dass*
 Maria weiß nicht, [*dass* Peter nach Hause *geht*]/ **[Peter geht nach Hause]*.
 Maria know not that Peter towards home go Peter go towards home
 (Intended:) 'Maria doesn't know that Peter is going home.'

I will refer to constructions that typically show up in root contexts as **root constructions**. German V2 clauses and English inverted questions, by this standard, are both root constructions.

Embedded root: The form-to-rootness mapping is not always neat; sometimes, root constructions show up in embedded clauses. Embedded root constructions are reported in a non-negligible number of languages:

- English embedded inverted questions (henceforth EIQs; McCloskey 2006; Dayal 2025)
- Hindi-Urdu polar question particle *kyaa* (Bhatt and Dayal 2020; Dayal 2025)
- Slovenian embedded imperatives (Kaufmann 2014; Stegovc and Kaufmann 2015; Kaufmann and Stegovc 2019)
- **German embedded V2 declaratives** (Reis 1997; Gärtner 2002; Truckenbrodt 2006)

Compared to their non-root counterparts, embedded root constructions are more distributively restricted:

- (8) *Believe* but not *doubt* can embed V2 clauses:
 Maria {*glaubt*/**bezweifelt*}, Peter *geht* nach Hause.
 Maria believe/doubt Peter go towards home
 (Intended:) 'Maria believes/doubts that Peter is going home.'
- (9) *Dass*-clauses embed under both *believe* and *doubt*:
 Maria {*glaubt*/*bezweifelt*}, *dass* Peter nach Hause *geht*.
 Maria believe/doubt that Peter towards home go
 'Maria believes/doubts that Peter is going home.'

The distributional restrictions on embedded root constructions form empirical and conceptual challenges to the subordination literature.

- Empirically, what affects the embedding profiles of these root constructions?
- Conceptually, where do such profiles arise from? What do embedded root constructions share at their hearts?
- The current study is an attempt to tackle the German V2 puzzle from an interfacial perspective.

The ideas:

- There are more than one point in the left periphery for building sentence meaning (Dayal 2025);
- Root constructions reflect the presence of extra structures in the left periphery;

- (10) *The Dayalian interrogative left periphery:*

$$\underbrace{\text{S(peech)A(ct)P(hrase)}}_{\text{root constructions}} > \underbrace{\text{Persp(ective)P}}_{\text{non-root}} > \text{CP}$$

- The syntax and the semantics of this extra structure contribute together to the non-canonical embeddability profile of embedded root constructions.

Overview

- Based on my survey and the literature, I establish the following basic properties of EV2:
 - EV2 clauses are syntactically different from *dass*-clauses (e.g., no EV2 clausal subjects);
 - V2-embedding predicates crosscut lexical subcategories and share a common semantic component;
 - The embedding of V2 is shiftY, i.e., sensitive to matrix contexts such as negation.
- I argue for a Persp(ective)P(hrase) analysis, extending the Dayalian interrogative left periphery across another major clause type.
- Specifically, I propose that the contrast between **embedded V2 clauses** and **embedded *dass*-clauses** boils down to a contrast between the declarative **PerspP (PerspP_p; 12a)** and **CP (CP_[¬Q]; 12b)**.

(11) *The Dayalian declarative left periphery:*

$$\text{SAP}_{\text{ASSERT}} > \underbrace{\text{PerspP}_p}_{\text{V2}} > \underbrace{\text{CP}_{[-Q]}}_{\text{dass}}$$

- The PerspP_p , merged as the extended projection of the clause-typing $\text{CP}_{[-Q]}$,
 - (i) anchors the embedded p to a perspectival center via PRO in Spec PerspP_p (12b), and
 - (ii) introduces a not-at-issue requirement on the doxastic state of the perspectival center (13a).
The minimal component is spelled out in (13b).
- (12) a. Embedded *dass*-clause: $[[\text{MatrixCP John thinks } [\text{EmbeddedCP}_{[-Q]} \textit{dass} \dots \text{V-final position}]]]$
 b. Embedded V2 clause: $[[\text{MatrixCP John}_i \text{ thinks } [\text{EmbPerspP}_p \text{ PRO}_i \text{ V2-position } [\text{EmbCP}_{[-Q]} \dots]]]]]$
- (13) a. $[[\text{PerspP}]]^c = \lambda p \lambda x : \text{P-COMMIT}(p, x).p$
 b. $\text{P-COMMIT}(p, x) \rightarrow \Diamond \text{believe}(x, p)$
- (14) Summary: EV2 clauses and *dass*-clauses

Properties	EV2	<i>dass</i> -clauses
① Restricted to verb complements	✓	✗
② Predicate subcategorization	✓	✗
③ Shiftiness under negation	✓	✗
Syntactic status	PerspP _P	CP _[-Q]

Roadmap

- | | | | |
|-----|---------------------------------------|-----|---|
| \$1 | Introduction | \$4 | Derivation and comparison to other accounts |
| \$2 | Syntax and semantic properties of EV2 | \$5 | Embeddable root constructions as PerspP |
| \$3 | The PerspP proposal | \$6 | Conclusion |

2 Properties of EV2

2.1 Syntactic properties

EV2 clauses are not quotations.

- Indexicals: EV2 clauses do not shift indexicals, unlike quotations.

- (15) a. Der muss denken, ich **bin** bescheuert
 That must think I am stupid
 'He must think that I am stupid!' $\rightsquigarrow x$ thinks **the speaker** is stupid
 (Acquired from **swr.de**)
- b. Der muss denken, „**ich bin bescheuert**“
 That must think I am stupid
 'He must think, "I am stupid!"' $\rightsquigarrow x$ thinks **x** is stupid

- Variable binding: Matrix quantifiers can bind pronouns in EV2 clauses (Meinunger 2004);

- (16) [Jeder denkt]_i, [**er**_i **kommt** zu kurz].
 everybody thinks he comes to short
 'Everybody_i thinks he_i comes off worst.' (Meinunger 2004, 314)

- Verbs that do not take quotations can still embed EV2, e.g., *sich herausstellen* 'to emerge'.

- (17) a. In den Kommentaren stellte sich heraus, er war nicht der Einzige.
 in the comments stand self out he was not the only.one
 'It emerged in the comments that he wasn't the only one.' (Example acquired from **GameStar**)
- b. ??In den Kommentaren stellte sich heraus, „er war nicht der Einzige“.

EV2 clauses are syntactically different from *dass*-clauses (cf. Reis 1997).

- ① *Dass*-clauses can surface as subjects (18a), whereas EV2 clauses cannot (18b).

- (18) a. [**Dass** Peter nach Hause geht], macht mich traurig.
 that Peter towards home go make me sad
 'That Peter is going home makes me sad.'
- b. *[Peter **geht** nach Hause], macht mich traurig.
 Peter go towards home make me sad
 Intended: 'That Peter is going home makes me sad.'

- ② Only *dass*-clauses but not EV2 clausees can serve as complements of prepositions (Reis 1997).

- (19) a. Sie blieben zusammen bis [**dass** der Tod sie schied].
 They stay.PST together until that the death they separate.PST
 'They stayed together until death did them part.'
- b. *Sie blieben zusammen bis [der Tod **schied** sie].
 They stay.PST together until the death separate.PST they
 Intended: 'They stayed together until death did them part.' (Reis 1997, 140)

③ “Topicalization”³ can target *dass*-clauses but not *EV2* clauses (Reis 1997; Truckenbrodt 2006).

- (20) a. [*Dass* Peter nach Hause geht]_i glaubt Maria *t_i*.
 that Peter towards home go think Maria
 ‘Maria thinks that Peter is going home.’
 b. **[*Peter geht nach Hause]_i glaubt Maria *t_i*.
 Peter go towards home think Maria
 Intended: ‘Maria thinks that Peter is going home.’ (Truckenbrodt 2006, 279)

④ Conjunction patterns show that *EV2* and *dass*-clauses are syntactically different (21), assuming Coordination of Likes despite counterexamples (Chomsky 1957).⁴

- (21) a. Peter glaubt, es regnet morgen oder es schneit morgen.
 Peter believe it rain tomorrow or it snow tomorrow
 ‘Peter believes that it will rain tomorrow or it will snow tomorrow.’
 b. ??Peter glaubt, *dass* es morgen regnet oder es schneit morgen.
 Peter believe that it tomorrow rain or it snow tomorrow
 c. ??Peter glaubt, es regnet morgen oder *dass* es morgen schneit.
 Peter believe it rain tomorrow or that it tomorrow snow
 (Truckenbrodt 2006, 280)

3. As Keine (2020) notes, in the German literature, the term “topicalization” here refers to movements to the prefield, i.e., the preverbal position in a V2 clause. This is somewhat misleading because what shows up in the prefield does not necessarily have to be a topic. I adopt this term for the sake of convenience.

4. Conjunction tests for clauses seem to be quite reliable in German. For example, conjoining an embedded *dass*-clause and an embedded question is possible. The example is acquired from Krifka’s 2005 lecture notes.

- (i) Maria weiß, [[_{CP1-Q}] *dass* die Party stattfindet] und [_{CP1+Q}] wer kommt]
 Maria know that the party happen and who come
 ‘Maria knows that the party is happening and she knows who comes.’

Furthermore, conjunction between infinitival *zu*-clauses and finite *dass*-clauses is impossible, even if the verb can take either when they show up alone. This is reported both in Standard German (2) and Austrian German (3).

- (ii) Standard German:
 a. ...Unter der Prämisse, [*dass* sie dabei friedlich sind] und [*dass* der Schwanz nicht peitscht].
 under the premise at.which friendly to be and that the tail NEG lash
 ‘...Under the premise that they are friendly and that the tail is not lashing.’ *dass* + *dass*: ✓
 b. ??...Unter der Prämisse, [dabei friedlich zu sein] und [*dass* der Schwanz nicht peitscht].
 under the premise at.which friendly to be and that the tail NEG lash
 Intended: ‘... Under the premise that they are friendly and the tail is not lashing.’ Inf. + *dass*: ✗
 (Weber, Bildhauer, and Münzberg 2024, 80)

- (iii) Austrian German:
 a. Was_i hat ea angfanga [*t_i* zum repariern] und [*t_i* gern zum haben]
 what has he started to repair and like to have
 ‘What did he start to repair *x* and like *x*?’ Inf + Inf: ✓
 b. Was hat ea angfanga [*dass* a *t_i* repariert] und [(*dass* a) *t_i* gern hat]
 what has he started that he repairs and (that he) like has
 ‘What did he start to repair *x* and like *x*?’ *dass* + *dass*: ✓
 c.*?Was hat ea angfanga [*dass* a *t_i* repariert] und [*t_i* gern zum haben]
 what has he started that he repairs and like to have
 Intended: ‘What did he start to repair *x* and like *x*?’ *dass* + Inf: ✗
 d.*?Was hat ea angfanga [*t_i* zum repariern] und [*dass* a *t_i* gern hat]
 what has he started to repair and that he like has
 Intended: ‘What did he start to repair *x* and like *x*?’ Inf + *dass*: ✗
 (Lohninger 2026)

Upshot: V2 clauses are syntactically distinct from *dass*-clauses and are restricted to verb complements only.

2.2 Predicate subcategorization and shiftiness

Not all verbs that take *dass*-clauses can take EV2, and c-selection alone is not enough to derive the data pattern

- ① Compared to non-EV2 verbs, EV2 verbs have a common semantic component: **commitment** (Gärtner 2002; Truckenbrodt 2006; Woods 2016; Nederveen 2020).

(22) EV2 verbs:

- a. Doxastic attitude predicates:
glauben 'believe', *meinen* 'mean/think', *finden* 'think', *denken* 'think', *wissen* 'know'⁵;
- b. Certainty predicates:
es ist wahr/sicher/klar 'it is true/for sure/clear', *feststehen* 'be certain/fixed', *voraussetzen* 'assume/take it for granted that';
- c. Verbs of saying:
sagen 'say', *behaupten* 'claim', *erzählen* 'tell', *angeben* 'state', *vorschlagen* 'suggest';
- d. (Some) Verbs of preference:
es ist beste/besser/lieber 'it is better', *vorziehen* 'prefer'.

(23) Non-EV2 verbs:

bezweifeln 'doubt', *vergessen* 'forget', *wollen* 'want', *es ist (un)möglich* 'It is (im)possible', *es ist (un)wahrscheinlich* 'It is (un)likely', etc.

(Reis 1997; Truckenbrodt 2006; subcategories translated from Reis 1997)

- (24) ...[S]etting predicates of certainty aside for the moment: V2 predicates establish a belief, speech, or preference world, alternative to the actual world, for their subject (where one exists), in which the dependent proposition is claimed to be true.

(Reis 1997, 122; translation mine)

- ② EV2 verbs crosscut existing syntactic subcategories.

- Some anti-rogative verbs takes EV2 (e.g., *glauben* 'believe'), and some don't (e.g., *verursachen* 'cause').
- Some responsive verbs take EV2 (e.g., *bemerken* 'notice'), and some don't (e.g., *vergessen* 'forget').
- Some factive verbs arguably takes EV2 clauses (e.g., *wissen* 'know').
- *Es ist sicher* 'it is for sure' takes EV2 clauses, but *es ist möglich* 'it is possible' doesn't.

- ③ Interestingly, the embedding profile of V2 are shifty, i.e., sensitive to the matrix.

- When negated, EV2 verbs cannot take V2 complements (Reis 1997; Meinunger 2004; Truckenbrodt 2006);

5. Reis (1997) observes that factive verbs, including *wissen* 'know', do not embed V2 clauses. However, Truckenbrodt (2006) argues that at least for *wissen*, the sentence is acceptable under a half-statement reading – a reading where the speaker also commits to the V2 clause. While this relates to the interesting issue of *de se* attitude reported in Dayal (2025) for English embedded questions, which we will return to later, for now, all we have to know is that the V2 clause is indeed embedded, as shown by the variable binding tests below:

- (i) [Jeder Spieler]_i weiß: er_i wird gebraucht.
every player know he is.PASS needed
'[Every player]_i knows, he_i is needed.'

(Example acquired from [ksc.de](https://www.ksc.de))

(25) *Hans glaubt nicht, Peter **geht** nach Hause.

Hans believe NEG Peter go towards home.

Intended: 'Hans doesn't believe Peter is going home.'

(Truckenbrodt 2006, 295)

- Infinitival adjuncts that indicate the attitude holder's disbelief also ban EV2 (26). Note that it is not the case that any infinitival adjunct can block EV2 (27).

(26) Wilhelm behauptet, (*ohne es zu glauben,) er **verdient** eine Beförderung.

William claim without it to believe he deserve a promotion

'William claims (*without believing) that he deserves a promotion.'

(27) Wilhelm behauptet, (und glaubt es selbst,) er **verdient** eine Beförderung.

William claim and believe it self he deserve a promotion

'William claims (, and believes,) that he deserves a promotion.'

- Despite cross-speaker variation, there seems to be a case where a non-EV2 verb embeds V2 clauses when negated (28);

(28) %Anton bezweifelt nicht: Clara **ist** vegetarisch.

Anton doubt NEG Clara is vegetarian

'Anton doesn't doubt that Clara is vegetarian.'

- ④ Modals, interrogative force, or imperative force does not affect the embedding profile;

(29) Modalized EV2 verbs still embed V2:

a. Der muss denken, ich **bin** bescheuert

That must think I am stupid

'He must think that I am stupid'

(Acquired from *swr.de*)

b. Man könnte denken, es **ist** sein Pokerface, aber ein Mann wie er pokert nicht.

people might think it is his pokerface but a man like he play.poker NEG

'One might think it is his poker face, but a man like him doesn't play poker.'

(Acquired from *Black Hearts*, by Geneva Lee)

(30) Imperatives:

Glauben Sie mir doch, ich **bin** unschuldig!

believe.IMP you.HON me.DAT EMPH I am innocent

'Please do believe me that I am innocent!'

(Acquired from *grammis*)

(31) Interrogatives:

a. Wen_i hat sie gesagt [_{*t_i*} **hat** der Peter _{*t_i*} getroffen](?)

who.ACC has she said has the Peter.NOM met

'Who did she say that Peter met?'

(Keine 2020, 220)

b. Wer glaubt, Peter **geht** nach Hause?

who believes Peter go towards home

'Who believes Peter is going home?'

(Truckenbrodt 2006, 296)

- In fact, modals, interrogative force, and imperative force seem to **cancel** the shifting effect of matrix negation.⁶

6. One complication is that the EV2 clauses in these examples are all in subjunctive mood, as noted by Meinunger (2004). Switching the EV2 to indicative mood in these cases will lead to unacceptability. Clearly, an account for the mood contrast relies on pinning down

- (32) Modals render negation ineffective:
- Freilich solltest du nicht denken, dieser gute Glaube sei eine Entschuldigung oder...
Surely should you NEG think this good faith be an apology or...
'Of course: you should not think that this belief would be an apology'
 - Sie dürfen nicht glauben, das Gericht hätte leichtfertig entschieden.
you.HON may not believe the court had carelessly decided
'You should not believe the court took the decision carelessly.'
(Meinunger 2004, 317–318; originally from Butolussi 1991)
- (33) Interrogatives render negation ineffective:
- Sagtest du nicht eben, die Doktrin,..., sei Ausdruck des Blutes?
say you NEG just the doctrine be expression of-the blood
'Didn't you just say the doctrine... was an expression of blood...?'
(Meinunger 2004, 318; originally from Butolussi 1991)
- (34) Imperative force renders negation ineffective:
- Glaub bloß nicht, ich helfe dir!
believe.IMP just NEG I help you
'Just don't believe I will help you!'
(Truckenbrodt 2006, 296)

Upshot: The subcategorization pattern and the shifty embeddability profile of V2 call for an interfacial account.

3 The PerspP proposal

I draw on Dayal's (2025) insight that embedded root structures involve extra projections (35) with not-at-issue semantic components.

- (35) *The Dayalian left periphery:*
- SAP > PerspP > CP
V2/IQ dass/whether

3.1 Parallelism between EV2 and EIQ

McCloskey (2006) for Irish English, as well as Woods (2016) and Dayal (2025) for Standard English, observes that inverted questions sometimes embed.

- Embedded inverted questions (EIQs) can only serve as complements of certain predicates (36a vs. 36b&36c) — recall my ① in §2.2 about subcategorization and ① in §2.1 about syntactic restrictions;
 - (36) a. Mary wonders, [was Henry a communist_↑].
 - b. *Mary remembers, [was Henry a communist_↑]
 - c. *[Was Henry a communist_↑] is important.
- EIQ verbs crosscut the rogative/responsive distinction: *forget* is a responsive verb that takes EIQs — this corresponds to my ②.

(37) I have forgotten, [did Ann get A's in her 1st year courses_↑].

the semantic contribution of mood, which is beyond the scope of the current study. For now, I do not distinguish subjunctive EV2 and indicative EV2 and analyze them in the same way.

- iii Matrix negation affects the embedding profile. Compare (38) to (36b) — my ③ reflects the same type of shiftiness.

(38) Mary doesn't remember, [was Henry a communist_T]

- (39) Summary: EV2 and EIQ

Properties	EV2	<i>dass</i> -clauses	EIQ	<i>whether</i> -clauses
① Restricted to verb complements	✓	✗	✓	✗
② Predicate subcategorization	✓	✗	✓	✗
③ Shiftiness under negation	✓	✗	✓	✗
Syntactic status	PerspP _p	CP _[-Q]	PerspP _Q	CP _[+Q]

3.2 Dayal (2025)

The extra projection, PerspP_Q has two semantic contributions.

- First, a perspectival center is introduced via PRO in Spec PerspP_Q (40a), which gets bound by either the matrix subject or the actual speaker.
- Second, the Persp⁰ introduces a not-at-issue requirement on the epistemic state of the perspectival center (the p(otential)-activeness requirement; 40b).
- The minimal component of P-ACTIVENESS is spelled out in (40c): It should be possible that the perspectival center is ignorant of the answer to the embedded question.

(40) Dayal's left periphery: ... PerspP_Q > CP_[+Q] > ...

- [_{MatrixCP} Attitude holder_i... [_{PerspP_Q} PRO_i Persp⁰_Q [_{CP_[+Q]} ...]]]
- [[Persp⁰_Q]^c = $\lambda Q \lambda x : \text{P-ACTIVE}(Q, x).Q$
- $\forall x \forall Q [\text{P-ACTIVE}(Q, x) \rightarrow \Diamond \neg \text{know}(x, \text{Ans}(Q))]$

(Adapted from p.673)

Sample derivation:

- The lexical semantics of *know* itself contradicts the p-activeness requirement (42a), whereas those of *wonder* and *don't know* are compatible with it (42b).

- (41) a. *John knows* [Q] $\rightsquigarrow \neg \Diamond \neg \text{know}(J, \text{Ans}(Q)) \rightsquigarrow \text{EIQ} \text{ ✗}$
b. *John doesn't know* [Q] $\rightsquigarrow \Diamond \neg \text{know}(J, \text{Ans}(Q)) \rightsquigarrow \text{EIQ} \text{ ✓}$

- This also correctly predicts that *forget/not-remember* will allow but *remember/not-forget* will not allow EIQs.

- (42) a. *John knows* [Q] $\rightsquigarrow \neg \Diamond \neg \text{know}(J, \text{Ans}(Q)) \rightsquigarrow \text{EIQ} \text{ ✗}$
b. *John doesn't know* [Q] $\rightsquigarrow \Diamond \neg \text{know}(J, \text{Ans}(Q)) \rightsquigarrow \text{EIQ} \text{ ✓}$

- Note that this requirement should be checked late enough in the derivation for matrix negation to come into play. For concreteness, I will think of it as a *postsupposition* (Brasoveanu and Szabolcsi 2013). That is, it is checked after the expression is integrated into the context.

3.3 Extending to German EV2

- I propose that German V2 signals the presence of PerspP_p in the declarative left periphery (43).

$$(43) \quad [\text{PerspP}_p \text{ PRO Persp}_p^0 [\text{CP}_{[-Q]} \dots]]$$

- I posit that the finite verb in a V2 clause head moves from V^0 to C^0 and then to Persp_p^0 .
- PerspP_p introduces the perspectival center via PRO, and a not-at-issue requirement on its doxastic state (44a). Following Dayal (2025), the PRO can be bound by either the attitude holder or the actual speaker.
- The minimal component is shown in (44b): For V2 clauses to be embedded, it should be possible that the perspectival center believes the embedded proposition.

$$(44) \quad \begin{aligned} \text{a.} \quad & \llbracket \text{Persp}_p \rrbracket^c = \lambda p \lambda x : \text{P-COMMIT}(p, x).p \\ \text{b.} \quad & \text{P-COMMIT}(p, x) \rightarrow \Diamond \text{believe}(x, p) \end{aligned}$$

- This way, we capture why EV2 verbs seem to share a commitment component in their semantics:

$$(45) \quad \begin{aligned} \text{a.} \quad & \text{John thinks } [p] \rightsquigarrow \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2 } \checkmark \\ \text{b.} \quad & \text{John doubts } [p] \rightsquigarrow \neg \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2 } \times \end{aligned}$$

- Why $\Diamond \text{believe}(x, p)$ instead of $\text{believe}(x, p)$ (cf. Truckenbrodt 2006)? We will come back to it in the next section.
- Why is the belief requirement a minimal requirement? We will come back to this in §5.

4 Deriving EV2 with the PerspP account

4.1 Derivations

What syntax gives us:

- §2.1 ①–③:
 - V2 clauses cannot show up as clausal subjects or objects of prepositions.
 - An EV2 clause cannot be fronted into the prefield of the matrix clause.
 → Assuming that clausal subjects, objects of prepositions, and the prefield in general has a clause size restriction (i.e., no larger than CP), the distributional contrast follows.
- §2.1 ④:
 - EV2 clauses can't be coordinated with *dass*-clauses.
 → Assuming Coordination of Likes, this naturally follows from the fact that PerspP and CP are different in size.
 - A side-note: Coordination of Likes in German, at least for clauses, seems to work well (see fn.4).

What semantics gives us:

- §2.2 ① & ②: The subcategorization pattern follows from the P-COMMIT requirement and the lexical semantics of the predicates (46).

$$(46) \quad \begin{aligned} \text{a.} \quad & \text{John thinks } [p] \rightsquigarrow \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2 } \checkmark \\ \text{b.} \quad & \text{John doubts } [p] \rightsquigarrow \neg \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2 } \times \end{aligned}$$

- §2.2 ③: The shiftiness effect of negation follows from the P-COMMIT requirement, and the fact that this requirement is checked as a postsupposition (47) & (48).

$$(47) \quad \begin{aligned} \text{a.} \quad & \text{John doesn't think } [p] \rightsquigarrow \neg \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2 } \times \\ \text{b.} \quad & \text{John doesn't doubt } [p] \rightsquigarrow \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2 } \checkmark \end{aligned}$$

- (48) a. *John claims* $[p] \rightsquigarrow \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2} \checkmark$
 b. *John claims without believing* $[p] \rightsquigarrow \neg \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2} \times$
 c. *John claims without doubting* $[p] \rightsquigarrow \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2} \checkmark$
- §2.2 ④: Modals (deontic modals at least)⁷, interrogatives, and imperatives do not affect EV2 because they keep both possibilities open. The perspectival center could either believe or not believe p .
- (49) a. *John should/may (not) believe* $[p] \rightsquigarrow \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2} \checkmark$
 b. *Do you (not) think* $[p]? \rightsquigarrow \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2} \checkmark$
 c. *Do (not) believe* $[p]! \rightsquigarrow \Diamond \text{believe}(J, p) \rightsquigarrow \text{EV2} \checkmark$
- One might wonder, how do we account for the fact that *es ist wahr* ‘it is true’ embeds V2 but *es ist nicht wahr* ‘it is not true’ and *es ist möglich* ‘it is likely’ don’t?
 - Recall that the actual speaker can bind the PRO as well!⁸
- (50) a. $[\text{MatrixSAP } \text{Sp}_i \dots [\text{MatrixCP It is true } [\text{PerspP}_p \text{ PRO}_i [p]]]]$
 $\rightsquigarrow \text{believe}(\text{Sp}, p) \rightsquigarrow \Diamond \text{believe}(\text{Sp}, p) \rightsquigarrow \text{EV2} \checkmark$
 b. $[\text{MatrixSAP } \text{Sp}_i \dots [\text{MatrixCP It is not true } [\text{PerspP}_p \text{ PRO}_i [p]]]]$
 $\rightsquigarrow \neg \Diamond \text{believe}(\text{Sp}, p) \rightsquigarrow \text{EV2} \times$
- What about *möglich* ‘likely’? Treating p-COMMIT as a postsupposition allows us to check it after pragmatic inferences are computed — which is exactly what we need.
 By choosing to commit to $\Diamond p$, a proposition weaker than p , the speaker indicates their lack of belief in p , which violates the p-commitment requirement (51).
- (51) $[\text{MatrixSAP } \text{Sp}_i \dots [\text{MatrixCP It is likely } [\text{PerspP}_p \text{ PRO}_i [p]]]]$
 $\rightsquigarrow [\text{MatrixSAP } \text{Sp}_i \dots [\text{It is only likely } [\text{PerspP}_p \text{ PRO}_i [p]]]]$
 $\rightsquigarrow \text{The alternative set of Sp's commitment: } \{\Box p, p, \Diamond p\}$
 $\rightsquigarrow \neg \Diamond \text{believe}(\text{Sp}, p) \rightsquigarrow \text{EV2} \times$

Upshot: Clause-size restrictions and the semantic requirement on the attitude holder’s possible commitment derives what we’ve seen.

4.2 Brief comparison with existing alternatives

- Gärtner (2002) and Truckenbrodt (2006): Proto-assertive force and absorption.
 - V2 carries a proto-assertive force that has to be anchored to the actual speaker or an attitude holder;
 - The proto-assertive force encodes the anchor’s commitment in the embedded proposition.
 - My modification: commitment ($\text{believe}(x, p)$) is too strong, it rules out acceptable cases that are predicted by the potential commitment ($\Diamond \text{believe}(x, p)$) account.
- Meinunger (2006): Double assertion.
 - The EV2 clauses are interpreted twice: once in its embedded position, and once in the scope of the matrix assertion operator.

7. One can imagine how complicate the interactions between epistemic modals and the p-COMMIT requirement might be. Predictions vary depending on whether factivity is a component of epistemic necessity modals. I have not enough data to draw any conclusion, neither on the embedding facts nor on the behavior of German epistemic modals.

8. I take the actual speaker to be represented in the matrix syntax (in SAP; cf. Speas and Tenny 2003), but the reader should feel free to insert their favorite theory for representing the speaker linguistically.

- That is, the speaker asserts both the attitude report and the embedded V2 clause.
 - Therefore, (i) the speaker has to commit to the embedded proposition (see also Woods 2024), and (ii) the embedded proposition has to be discourse-new.
 - While this account captures an important intuition we will talk about soon, it undergenerates (52)–(54).
- (52) Sie sagt, es **wird** morgen regnen.
 she say it will tomorrow rain
 ‘She says it will rain tomorrow.’
 ~> Fine when the speaker does not believe it will rain tomorrow (Keine 2020, 223 fn.10)
- (53) [Jeder denkt]_i, er_i **kommt** zu kurz.
 everybody thinks he comes to short
 ‘Everybody_i thinks he_i comes off worst.’ (Meinunger 2004, 314)
- (54) a. ...Oder glaubst du, ich **bin** blöd?
 or think you I am stupid
 ‘Or do you think I am stupid?’ (Acquired from goethe.de)
- b. Der muss denken, ich **bin** bescheuert
 That must think I am stupid
 ‘He must think that I am stupid!’ (Acquired from swr.de)

4.3 Discourse updates, half-statement, and attitude *de se*

- The double assertion account reflects an important intuition/puzzle — both discussed in Truckenbrodt (2006) and Truckenbrodt and Sode (2020) and Dayal (2025) — about who holds the attitude *de se* in embedded root constructions.
 - Dayal (2025): Question *de se*
 Sometimes even if the question is embedded (55a), the actual addressee is obliged to answer it, as if the speaker is asking the question (55b).
- (55) (*Department secretary to new faculty member:*)
 a. [The chair wants to know [[can you teach Semantics I next semester?](#)]]
 b. (*Faculty member:*) Yes, I can./#Great that she cares. (Adapted from Dayal 2025, 685)
- Truckenbrodt (2006): Half-statement
 Some EV2 predicates give rise to a *half-statement* reading, where the actual speaker seems to assert the embedded clause as well.
- (56) Hans weiß, Peter **geht** nach Hause.
 Hans know Peter go towards home
 ‘Hans knows that Peter is going home.’
 ~> The speaker also believes that Peter is going home. (Truckenbrodt 2006, 299)
- This could be a general property of embedded PerspP (cf. Dayal and Grimshaw 2009). But why do we see it only in some cases but not others?
 - To me, there are at least three sub-cases that can be explained:
 - (i) Where there is no overt attitude holder in the matrix, the speaker binds PRO as the last resort (e.g., *the question is...* in English, German *es ist wahr* ‘it is true...’, etc.);

- (ii) Where the embedding predicate carries a presupposition about the speaker (e.g., German factive verbs like *wissen* ‘know’);
- (iii) Where PRO is anchored to the matrix argument but the whole utterance is treated as an indirect discourse move. The discourse update is a result of the interlocutors’ pragmatic reasoning process.
- If all update effects fall into the three kinds of cases above, we might not need to tie embedded root constructions to direct discourse updates. I leave this for future studies.

5 Zooming out: What’s in a PerspP?

Conjecture: A PerspP inherits (some of) the felicity conditions of the matrix discourse move of its type

- Now that I have argued for the PerspP_{*p*}, let us take a step back and think about what PerspP shares in common.
- Specifically, let’s take a look at the felicity conditions of questioning (57) and asserting (58). The underlined condition is what Dayal (2025) and this talk argue to be the not-at-issue requirements of PerspP.

(57) When the speaker utters a question (with a radical *p*?), the following requirements should be met.

- a. Speaker’s ignorance:
The speaker does not know the truth about *p*.
- b. Speaker’s interest:
The speaker wants to know the truth about *p*.
- c. Addressee’s competence:
The speaker herself believes that the addressee knows the truth about *p*. (Farkas 2024, 11)

(58) When the speaker asserts *p*, the following requirements should be met.

- a. Open issue: The speaker assumes that the issue *P* that her move places on the Table is open in *c_i*, i.e., *P* is not resolved in *c* but is resolvable in some future context *c'* monotonically reachable from *c*.
- b. Ad responder: The participant who is assumed to respond to the speech act is the addressee.
- c. Sp competence: Where *p* is the unique alternative in *P*, Sp presents herself as believing that $w_{@} \in p$, and as having some reason to support this belief.
- d. Ad ignorance: The speaker assumes that the addressee does not believe that $w_{@} \in p$.
- e. Ad compliance: The speaker assumes that the addressee will accept *p* in the future of the conversation. (Farkas 2024, 10)

- For EIQs, the not-at-issue requirement seems to be more than just ignorance (Dayal 2025):

(59) The interest requirement seems to be present

- a. Speaker A is writing annual evaluation letters for graduate students. She asks her colleague (43a): I have forgotten, [_{PerspP} did Ann get A’s in her 1st year courses]?
- b. Speaker A: Did Ann get A’s in her 1st year courses?
Speaker B: I used to know but now I’ve forgotten
[_{CP} whether she did/whether she got A’s in her 1st year courses].
*[_{PerspP} did she↑ / did she get A’s in her 1st year courses↑].

(683)

- For now, I remain agnostic about what other components are present in PerspP_{*p*}.

- Based on the observations on German embedded imperatives (Kaufmann and Poschmann 2013) and Slovenian embedded (Stegovec and Kaufmann 2015), I claim without establishing that there is an imperative counterpart, i.e., $\text{PerspP}_{\text{IMP}}$.
- Put together, I make the conjecture in (60).

(60) *The Embedded Root Conjecture:*

- a. Embeddable root constructions are exponents of PerspP; Embedded root constructions are embedded PerspPs;
- b. PerspP introduces some (but not all) felicity conditions for canonical utterances, leading to the restricted embeddability of embedded root constructions.

6 Conclusion

Extending Dayal's analysis to the declarative left periphery and providing an explanation for restrictions on German embedded V2 lends further support for a semantically sensitive articulated left periphery. The study also indicates a syntactic-semantic link between embeddable root constructions and their position in the left periphery; if a "root" construction is built at PerspP, we expect it to be restrictedly embed, with a limited possibility space of the restrictions.

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Appendix A: Against a complementation/modification account

- Reis (1997) and following studies argue that the syntactic contrast resembles that of complementation and modification. The idea is that *dass*-clauses are genuine complements, whereas EV2 clauses are, grossly simplifying, adjuncts.
 - While this contrast is attested in many languages, I argue that such an analysis does not apply to our German data.
 - To begin with, we never see the complement (*dass*-clauses) and the adjunct (EV2 clauses) spelled out at the same time.
 - Also, there is no positive evidence for seeing EV2 clauses as adjuncts.
 - * Lack of complement selection (e.g., Bochnak and Hanink 2022): not attested;
!! In fact, all EV2 verbs also embeds *dass*-clauses.
 - * Overt *say*-coverb (e.g., Major 2024): not attested;
 - * Multiple interpretations co-varying with attachment site (e.g.): not attested;
 - * Factive/non-factive contrast (e.g., Angelopoulos 2026): not attested.
 - This is still under debate, but extraction from embedded V2 clauses is argued to be possible (Keine 2020). Since I believe in Adjunct Islands, this does seem to undermine the adjunct analysis.
- (61) {Wen_i/Den Martin_i} hat sie gesagt [*t_i* hat der Peter *t_i* getroffen](?)
 who.ACC the Martin.ACC has she said has the Peter.NOM
 ‘Who did she say that Peter met?’/‘Martin, she said that Peter met.’ (Keine 2020, 220)
- In the absence of positive evidence, and the presence of tentative negative evidence, I adopt the PerspP-complement approach for the current study.

(Unfinished) Appendix B: EV2 with *dass*?

An insightful reviewer brought up an interesting study on colloquial German. It seems that sometimes we see embedded V2 clauses cum complementizers (Freywald 2008). This is unexpected in the current account (and in the V-to-C movement literature), because the presence of an overt complementizer blocks the movement from V⁰. I will try and think of something to say about this.